

PROJECT DESIGN PHASE

Metro Ticket Generating System in ServiceNow

Project Design Phase

This phase translates the approved requirements into a practical ServiceNow solution design.

1. Problem – Solution Fit

Problem–solution fit is established by matching recurring passenger pain points with the proposed ServiceNow capabilities.

Customer Problem	Observed Need	Proposed Capability	Fit / Benefit
Ticket booking can be slow or manual	Fast digital booking	Service Catalog item	Reduces booking effort
Fare calculation can be confusing	Accurate fare before confirmation	Client Script fare matrix	Automates calculation
Payment choices vary	Flexible payment options	UPI / Card / Others	Supports different users
Digital ticket access is desirable	Easy ticket presentation	QR widget	Provides digital QR representation
Operational records need consistency	Centralized ticket information	Metro Database + Flow Designer	Automates record creation
Request IDs can be difficult to read	Readable tracking	RITM Number field	Improves operational traceability

2. Proposed Solution

S.No.	Parameter	Description
1	Problem Statement	Metro passengers need a simple, fast and reliable digital process to book tickets with accurate fares and an accessible digital ticket.
2	Idea / Solution Description	Build a ServiceNow Metro Ticket Generating System using Service Catalog, station master data, client-side fare logic, payment UI policy, QR widget and Flow Designer.
3	Novelty / Uniqueness	Combines ticket booking, route-based fare automation, conditional payment details,

		QR presentation and ServiceNow transaction management in one workflow.
4	Social Impact / Customer Satisfaction	Reduces waiting/manual effort, improves convenience, supports cashless options and reduces paper dependency.
5	Business Model (Revenue Model)	The project is primarily a service automation solution. In a production metro environment, it can support fare collection and operational efficiency; no specific revenue model was defined in the supplied project brief.
6	Scalability of the Solution	Additional stations, fare rules, payment integrations, reports, notifications and production ticket validation can be added without redesigning the overall workflow.

3. Solution Architecture

Architecture and data flow

Layer	Component	Responsibility
User Layer	Passenger	Provides route, journey, passenger and payment selections.
Experience Layer	Service Portal / Service Catalog	Displays Book A Metro Ticket form and confirmation.
Logic Layer	Catalog Client Script	Validates route and calculates fare.
Logic Layer	Catalog UI Policy	Controls visibility of Enter Payment Mode.
Digital Ticket Layer	Metro QR Widget + QR API	Displays QR representation of the ticket.
Automation Layer	Flow Designer — Metro Project	Reads catalog variables and creates the Metro Database record.
Data Layer	Metro Station's Details	Stores station master records.
Data Layer	Metro Database	Stores ticket number, RITM, route, passenger, fare, payment and QR status.
Security Layer	ServiceNow ACL / Roles	Controls access to custom data.

Architecture Flow

**Passenger → Service Catalog → Client Script Fare Logic → Payment/UI Policy → QR Ticket →
Flow Designer → Metro Database**

Key design decisions:

- Metro Station's Details is used as reference data for Starting From and Going To.
- Fare calculation is handled on the catalog form using the configured station-to-station fare matrix.
- Request Item remains a reference to sc_req_item; a separate RITM Number string stores the readable RITM number.
- Metro Database uses ServiceNow auto-numbering with prefix MTKT and five digits.
- Flow Designer maps catalog variables into the Metro Database and maps Requested Item → Number into RITM Number.
- QR generation is presented through a ServiceNow widget and QR image URL.